



Product Service

CERTIFICATE

No. Z2 110557 0003 Rev. 00

Holder of Certificate: **MINDRA GREEN ENERGY LLP**
102, Devraj Industrial Park, OFF, Pirana Piplaj Road
Piplaj
Ahmedabad, Gujarat 382405
INDIA

Certification Mark:



Product: **Converter**
(PV grid-interactive inverter)

The product was tested on a voluntary basis and complies with the essential requirements. The certification mark shown above can be affixed on the product. It is not permitted to alter the certification mark in any way. In addition, the certification holder must not transfer the certificate to third parties. This certificate is valid until the listed date, unless it is cancelled earlier. All applicable requirements of the testing and certification regulations of TÜV SÜD Group have to be complied. For details see: www.tuvsud.com/ps-cert

Test report no.: 64290160004002D

Valid until: 2022-12-20

Date, 2020-12-21

(Billy Qiu)

CERTIFICATE

No. Z2 110557 0003 Rev. 00

Model(s): MIND 10K, MIND 12K, MIND 15K, MIND 17K, MIND 20K

Parameters:

Model:	MIND 10K	MIND 12K	MIND 15K	MIND 17K	MIND 20K
Vmax PV (Vd.c.)	1000	1000	1000	1000	1000
Isc PV (Ad.c)	15x2	15x2	23x2	23x2	23x2
Maximum continuous a.c. output current(Aa.c)	15	18	22	25	29
Maximum continuous a.c. output power(kW)	10	12	15	17	20
Nominal AC voltage (Va.c.)	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400
Nominal a.c. output frequency (Hz)	50/60	50/60	50/60	50/60	50/60
Protection class	I	I	I	I	I
IP code	IP65	IP65	IP65	IP65	IP65

Tested according to:

IEC 62109-1(ed.1)
IEC 62109-2(ed.1)
IEC 61727(ed.2)
IEC 62116(ed.2)

Production Facility(ies):

076189



Product Service

CERTIFICATE

No. Z2 110557 0001 Rev. 00

Holder of Certificate: **MINDRA GREEN ENERGY LLP**
102, Devraj Industrial Park, OFF, Pirana Piplaj Road
Piplaj
Ahmedabad, Gujarat 382405
INDIA

Certification Mark:



Product: **Converter**
(PV grid-interactive inverter)

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Test report no.: 64290160003702C

Valid until: 2022-12-20

Date, 2020-12-21

(Billy Qiu)



CERTIFICATE

No. Z2 110557 0001 Rev. 00

Model(s): MIND 10K, MIND 12K, MIND 15K, MIND 17K, MIND 20K

Parameters:

Model:	MIND 10K	MIND 12K	MIND 15K	MIND 17K	MIND 20K
Vmax PV (Vd.c.)	1000	1000	1000	1000	1000
Isc PV (Ad.c)	29	29	46	46	46
Maximum continuous a.c. output current(Aa.c)	15	18	22	25	29
Maximum continuous a.c. output power(kW)	10	12	15	17	20
Nominal AC voltage (Va.c.)	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400
Nominal a.c. output frequency (Hz)	50/60	50/60	50/60	50/60	50/60
Protection class	I	I	I	I	I
IP code	IP65	IP65	IP65	IP65	IP65

Tested according to:

IEC 62109-1(ed.1)
EN 62109-1:2010
IEC 62109-2:2011
EN 62109-2:2011

Production Facility(ies):

076189



Product Service

Attestation of Conformity

No. N8A 110557 0002 Rev. 00

Holder of Certificate: **MINDRA GREEN ENERGY LLP**
102, Devraj Industrial Park, OFF, Pirana Piplaj Road
Piplaj
Ahmedabad, Gujarat 382405
INDIA

Product: **Converter**
(PV grid-interactive inverter)

This Attestation of Conformity is issued on a voluntary basis according to the Low Voltage Directive 2014/35/EU relating to electrical equipment designed for use within certain voltage limits. It confirms that the listed equipment complies with the principal protection requirements of the directive and is based on the technical specifications applicable at the time of issuance. It refers only to the particular sample submitted for testing and certification. For details see: www.tuvsud.com/ps-cert

Test report no.: 64290160003702C

Date, 2020-12-21

(Billy Qiu)

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After preparation of the necessary technical documentation as well as the EU declaration of conformity the required CE marking can be affixed on the product. The declaration of conformity is issued under the sole responsibility of the manufacturer. Other relevant EU-directives have to be observed.



Product Service

Attestation of Conformity

No. N8A 110557 0002 Rev. 00

Model(s): MIND 10K, MIND 12K, MIND 15K, MIND 17K, MIND 20K

Parameters:

Model:	MIND 10K	MIND 12K	MIND 15K	MIND 17K	MIND 20K
Vmax PV (Vd.c.)	1000	1000	1000	1000	1000
Isc PV (Ad.c)	29	29	46	46	46
Maximum continuous a.c. output current(Aa.c)	15	18	22	25	29
Maximum continuous a.c. output power(kW)	10	12	15	17	20
Nominal AC voltage (Va.c.)	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400	3/N/PE 230/400
Nominal a.c. output frequency (Hz)	50/60	50/60	50/60	50/60	50/60
Protection class	I	I	I	I	I
IP code	IP65	IP65	IP65	IP65	IP65

Tested according to:

EN 62109-1:2010
EN 62109-2:2011



Product Service

Test Verification

Dated: 2020-11-06

Applicant: MINDRA GREEN ENERGY LLP
102, Devraj Industrial Park, OFF, Pirana Piplaj Road,
Piplaj Ahmedabad, Gujarat 382405, INDIA

Product: PV Grid-interactive Inverter

Model: MIND 10K, MIND 12K, MIND 15K, MIND 17K,
MIND 20K

Tested according to: IEC 61683:1999;
IEC 60068-2-1:2007;
IEC 60068-2-2:2007;
IEC 60068-2-14:2009;
IEC 60068-2-30:2005;

Test report No.: 64.290.16.00039.01D

TÜV SÜD Certification and Testing (China) Co., Ltd. Guangzhou Branch

Section Manager
[Billy Qiu]

This Verification confirms that the test results of the presented product in the above listed test report are in compliance with the specified requirements. This Verification is part of the full test report(s) and should be read in conjunction with it.



Product Service

Attestation of Conformity

No. E8A 110557 0011 Rev. 00

Holder of Certificate: **MINDRA GREEN ENERGY LLP**
102, Devraj Industrial Park, OFF, Pirana Piplaj Road
Piplaj
Ahmedabad, Gujarat 382405
INDIA

Name of Object: **Converter**
(PV grid-interactive inverter)

This Attestation of Conformity is issued on a voluntary basis according to the Directive 2014/30/EU relating to electromagnetic compatibility. It confirms that the listed apparatus complies with all essential requirements of the directive and is based on the technical specifications applicable at the time of issuance. It refers only to the particular sample submitted for testing and certification. For details see: www.tuvsud.com/ps-cert

Test report no.: 687721600201F

Date, 2020-12-28

(Laurent Yuan)

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After preparation of the necessary technical documentation as well as the EU Declaration of conformity the required CE marking can be affixed on the product. That Declaration of conformity is issued under the sole responsibility of the manufacturer. Other relevant EU-directives have to be observed.



Product Service

Attestation of Conformity

No. E8A 110557 0011 Rev. 00

Model(s): MIND 10K, MIND 12K, MIND 15K,
MIND 17K, MIND 20K

Description of Object:

Rated Input: See below table for details

Model:	MIND 10K	MIND 12K	MIND 15K	MIND 17K	MIND 20K
V _{max} PV (Vd.c.)	1000	1000	1000	1000	1000
MPPT voltage range (Vd.c.)	480-800	480-800	480-800	480-800	480-800
Maximum operating PV input current (Ad.c.)	13x2	13x2	21x2	21x2	21x2
I _{sc} PV (Ad.c.)	15x2	15x2	23x2	23x2	23x2
a.c. output voltage (Va.c.)	3/N/PE, 400/230V	3/N/PE, 400/230V	3/N/PE, 400/230V	3/N/PE, 400/230V	3/N/PE, 400/230V
Nominal a.c. output frequency (Hz)	50	50	50	50	50
Maximum continuous a.c. output current(Aa.c.)	15	18	22	25	29
Maximum continuous a.c. output power(kW)	10	12	15	17	20
Power factor range	0.8 under-excited to 0.8 over-excited	0.8 under-excited to 0.8 over-excited	0.8 under-excited to 0.8 over-excited	0.8 under-excited to 0.8 over-excited	0.8 under-excited to 0.8 over-excited
Protection class	I	I	I	I	I
IP code	IP65	IP65	IP65	IP65	IP65

Tested according to: EN 61000-6-4:2007/A1:2011
EN 61000-6-2:2005

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After preparation of the necessary technical documentation as well as the EU Declaration of conformity the required CE marking can be affixed on the product. That Declaration of conformity is issued under the sole responsibility of the manufacturer. Other relevant EU-directives have to be observed.